Lab Description

This lab introduces students to the core principles of sensors and actuators within water level control systems. **Sensors** detect physical properties like temperature, pressure, flow rate, and weight, converting them into electrical signals. **Actuators** translate control signals into physical actions, such as adjusting motor speeds or valve positions. We begin by constructing a circuit to integrate sensors (scale) and actuators (pumps), followed by calibration. Through data collection and curve fitting, students will develop functions to convert between flow rate and signal for the pump and establish the relationship between weight readings and water volume for the scale. This hands-on experience enhances their understanding of experimental setup, data analysis, and function development.

Learning Outcomes

- Identify the physical components of process control schemes
- Generate calibration data for sensors and actuators
- Interpret calibration data for sensors and actuators

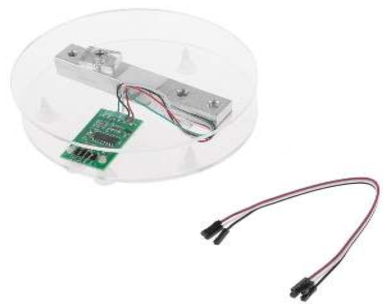
Sensor and Actuator Basics

Sensors: Devices that detect and measure physical properties (e.g., temperature, pressure, flow rate, Weight) and convert them into electrical signals for processing. Sensors provide feedback about the current state of a process.

Actuators: Devices that receive signals from a control system and convert them into physical actions or control signals to manipulate a process (e.g., opening or closing a valve, adjusting a motor speed).

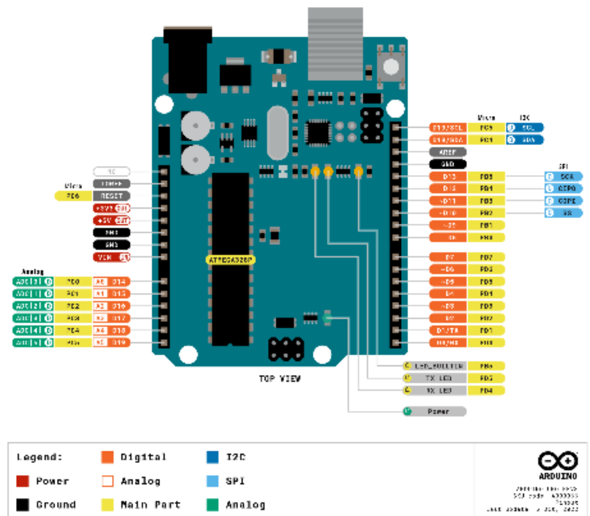
Sensors provide data to the control system, allowing it to monitor and adjust the process variables to maintain desired conditions or achieve specific goals. Actuators execute control actions based on signals from the control system, influencing the process to achieve the desired outcome. Together, sensors and actuators form the feedback loop essential for closed-loop control in process dynamics and control.

Materials Required



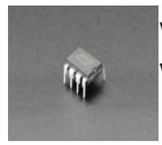
20kg Digital Load Cell Weight Sensor

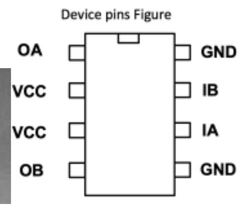
The 20kg Digital Load Cell Weight Sensor uses a strain gauge attached to a sturdy aluminum base to measure force or weight. When force is applied, the strain gauge's resistance changes, producing a different electrical output. Note that calibration with known weights is required after installation.



Arduino UNO

The Arduino UNO is a microcontroller board based on the ATmega328P microcontroller. It works by executing instructions stored in its memory, which are written in the Arduino programming language (based on C/C++). The board has digital and analog input/output pins that can be used to interface with various sensors and actuators. These pins can read digital or analog signals from sensors and output signals to control actuators. By writing code and uploading it to the Arduino UNO, users can create interactive electronic projects that respond to their environment.

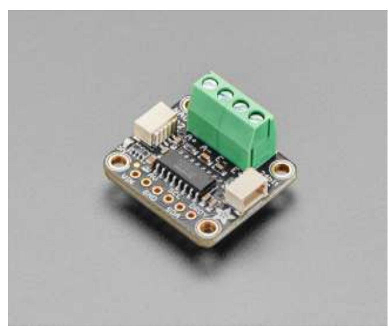




Symbol	Function
OA	A road output pin
VCC	Supply Voltage
VCC	Supply Voltage
OB	B output pin
GND	Ground
IA	A road input pin
IB	B input pin
GND	Ground

L9110H H-Bridge Motor Driver

The L9110H H-Bridge Motor Driver is a chip that can control the direction and speed of a DC motor. It works by using four transistors to control the flow of current through the motor. By applying different combinations of signals to its input pins, the L9110H can make the motor turn in forward, reverse, or stop it altogether. It is commonly used in robotics and other projects requiring motor control.



NAU7802 24-Bit ADC

A high-resolution sensor that can measure tiny changes in voltage. It's designed to work with sensors like strain gauges or load cells, which are often used to measure stress or strain. The ADC converts these voltage changes into digital values that can be read by a microcontroller, like an Arduino.

JST SH 4-Pin to Premium Male Headers Cable
A connector cable that allows you to easily connect Qwiic-compatible devices to your projects.

Jumper Wires
Used to make temporary connections in breadboards, circuits, or between components.

Half Sized Premium Breadboard with 400 tie points
A prototyping board used to build and test electronic circuits. It consists of interconnected metal clips housed in a plastic case, providing a platform for mounting and connecting electronic components without the need for soldering.

Submersible 3V DC Water Pump

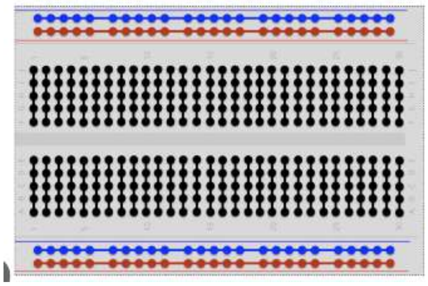
A small water pump designed to be submerged in water and operated using a 3V DC power source. Can pump water to a maximum of 1 meter high.

Others:

500 ml Beakers (x2), Water, and Tubing for Submersible Pumps (PVC 6mm ID)

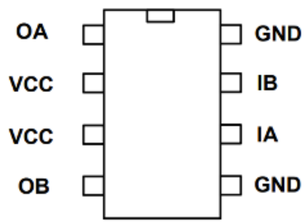
Using a Breadboard

Breadboards allow us to create more complex circuits than just connecting directly to the Arduino Uno. The pins of the breadboard are connected as shown below:



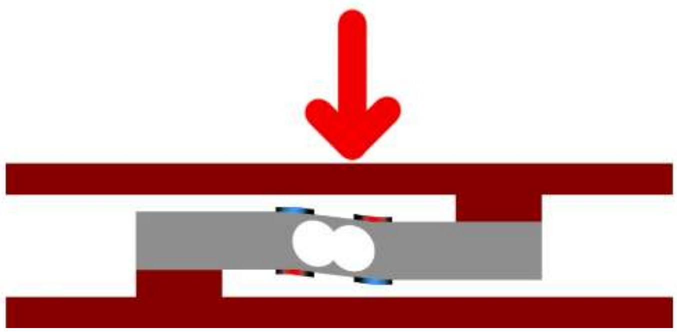
DC Motors and H-Bridges

The submersible pump is powered by a 3V DC motor. To be able to set the desired flowrate, a H-Bridge is required. The H-Bridge uses four switches arranged in a letter H. These four switches are used to control motor speed and direction (in the case of the submersible pump, there is only one direction).



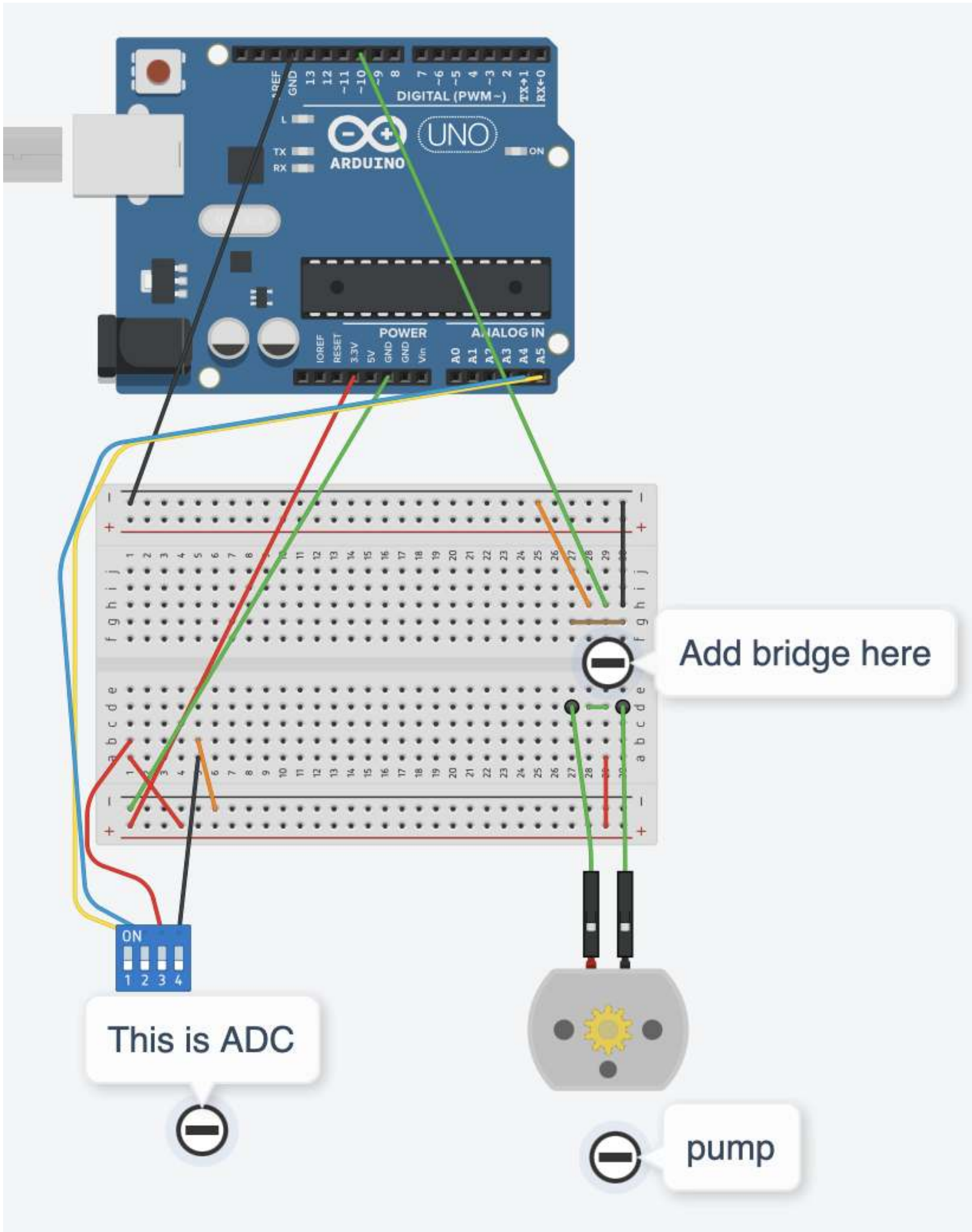
Load Cells

The 4 wire load cell deforms slightly under strain which results in a small change in the resistance. This small change in the resistance is measured and amplified by the ADC for measurements and analysis.

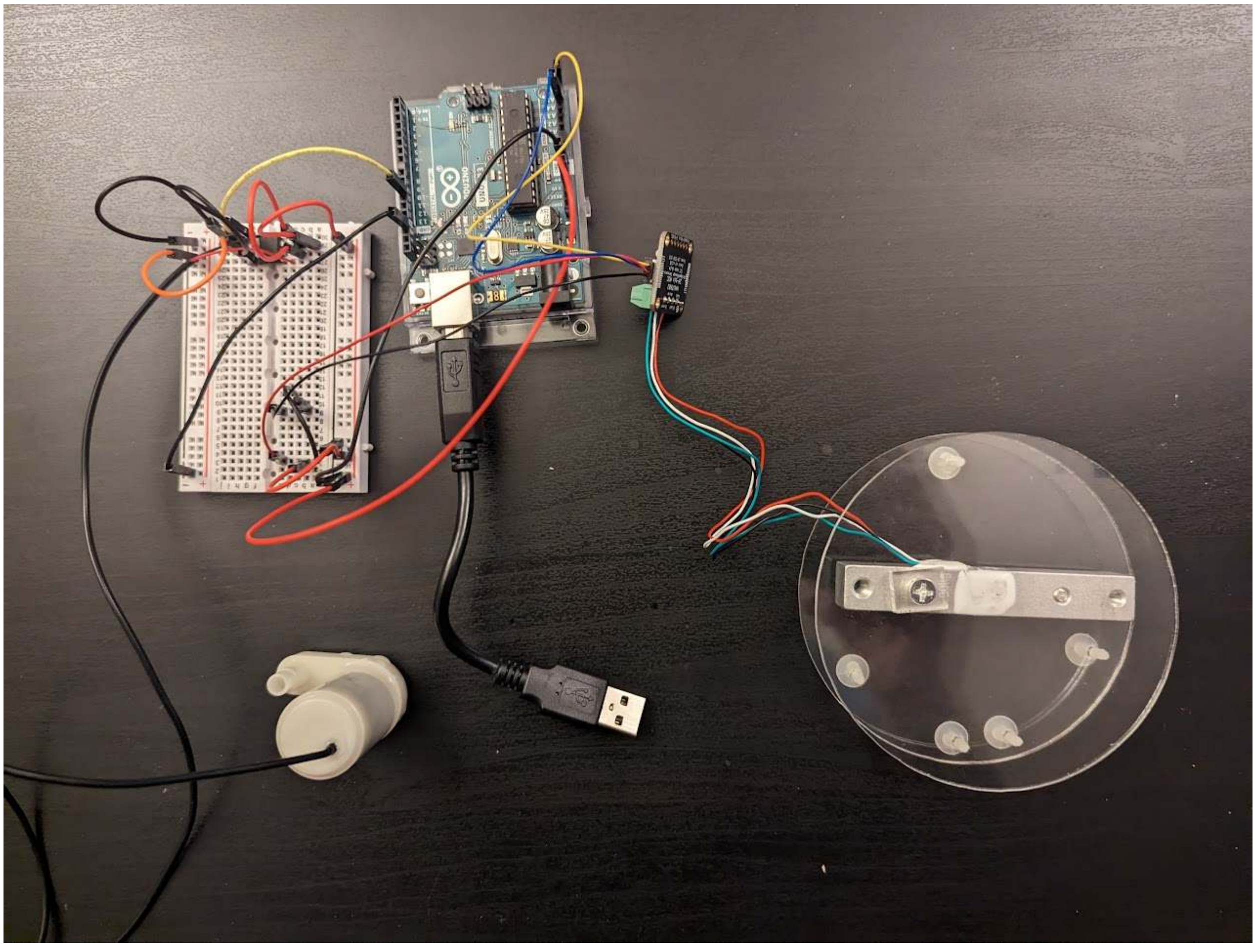


Building the Circuit

Use the circuit diagram below to connect the pump, H-Bridge, and scale to the Arduino Uno using the breadboard. Make sure the notch of the H-Bridge is pointing to the center of the board.



The completed circuit should look something like the one below:



Before proceeding, have one of the TAs check that the circuit is wired up correctly.

Code Required

The Ardunio Uno and Simulink communicate to one another using serial communication. The Arduino Uno does not know it is communicating with Simulink just as Simulink does not know it is communicating with the Arduino Uno. Serial communication works by just reading and writing a bit at a time. To accomplish this, you will need the Arduino sketch (code file) and Simulink simulation file. A brief description of each is given below:

Arduino Sketch:

- Read the value from the scale and write it to Simulink.

- Read the set point power percentage from Simulink to set the pump power
- Be sure to download the Adafruit NAU7802 Library in the Arduino IDE

MATLAB Simulink Simulation:

- Receive the scale value from Arduino for visualization and calibration.
- Output the power percentage for the pump to Arduino to further set the pump power.

To set up the serial communication, follow the steps below:

1. Download the Arduino sketch and MATLAB Simulink file.
2. Use the Arduino IDE to open the sketch.
3. Connect the Arduino Uno to your computer.
4. Check that the board is selected and make note of the COM#.
5. Upload the sketch to the Arduino Uno.
6. Open the MATLAB Simulink file
7. Change the COM# on the serial configuration, serial receive, and serial send blocks

Calibrating the Pump

1. Gather Materials:

- 500 mL beaker with 100 ml graduations
- Water
- Pump setup
- Notebook and pen for recording measurements

2. Prepare the Setup:

- Set up your submersible pump in a container filled with water. Ensure the pump is fully submerged.

3. Determine the Flow Rate:

- Start with the pump at 0% power (off).
- Set the pump power in Simulink to 50% and use a stopwatch to measure the time it takes for the pump to pump a certain volume of water.
- Record these values.

4. Repeat:

- Repeat the measurement at different power settings (e.g., increment of 10% i.e. 60,70,..,100%) and for the same volume of water.

5. Create a Calibration Curve:

- Plot a graph of percent power (x-axis) vs. flow rate (y-axis) to create a calibration curve.
- This curve can be used to determine the power level needed to achieve a specific flow rate of water.

Calibrating the Scale

1. Gather Materials:

- 500 mL beaker with 100 mL graduations
- Water
- Scale to be calibrated
- Notebook and pen for recording measurements

2. Record Empty weight of Scale:

- Place the empty 500 mL beaker on the scale to account for the weight of the beaker.
- Record the scale value.

3. Get Scale Values:

- Fill the beaker with water up to the 100 mL graduation mark.
- Record the scale value.

4. Repeat:

- Empty the beaker and repeat for each 100 mL increment (i.e., fill to 200 mL, 300 mL, 400 mL, and 500 mL) until you have recorded all six measurements.

5. Plot Data:

- Plot the recorded scale readings (y-axis) against the volume of water (x-axis) to create a calibration curve.
- This curve can be used to determine the volume level from the scale reading.

Submission

Have one team member submit both calibration curves and equations representing the curves.